

Professional Summary

Accomplished IT professional with 6+ years of experience in leading cross-functional diverse teams spread across multiple geographies. Influential and effective team leader with expertise in database administration, data engineering, database reliability engineering, data science, data governance, master data management, data security, project and program management, IT operations and large-scale database center migration and consolidations.

- Strong experience in Software Development Life Cycle (SDLC) including Requirements Analysis, Design Specification and Testing as per Cycle in both Waterfall and Agile methodologies.
- Hands-on experience in designing and implementing data engineering pipelines and analyzing data using AWS stack like AWS EMR, AWS Glue, EC2, AWS Lambda, Athena, Redshift, Scoop and Hives
- Strong experience in writing scripts using Python API, PySpark API and Tableau Spark API for analyzing the data. And extensively used Python Libraries PySpark, Pytest, Pymongo, coracle, PyExcel, Boto3, Pyscopg, EmbedPy, Power BI NumPy and Beautiful Soup.
- Experience in Google Cloud components, Google container builders and GCP client libraries and cloud SDK's
- Experience in setting up monitoring infrastructure for Hadoop cluster using Nagios and Ganglia.
- Working with AWS/GCP cloud using in GCP Cloud storage, Data-Proc, Data Flow, Big- Query, EMR, S3, Glacier and EC2 Instance with EMR cluster.
- Hands on strong experience in data modeling, design, and architecture on AWS and Azure; designing cloud-native data lakes and distributed data platforms, expertise in various scripting languages R, Python, JavaScript.
- Expertise in Python (2.x/3.x) programming with multiple packages including NumPy, Pandas, boto, boto3
- Sound experience with terraform Infrastructure/Configuration-as-code tooling.
- Expertise in Dock erization of application and configuring container orchestration. I.e. Kubernetes, Mesos, AWS ECS, Helm package manager and CI/CD.
- Experience with hybrid cloud deployments, and on-premises-to-cloud migration deployments and roadmaps.
- Good working knowledge of classes for centralizing all the frequently used methods, which are invoked in Lambda handlers to create and maintain custom libraries in Python.
- Adept in analyzing large datasets using Apache Spark, Spark ML and Amazon Web Services (AWS).
- Proficient in working with SQL/non-SQL database, AWS EC2, S3, EMR, Lambdas, Sage Maker, Docker.
- Experience in Data Governance consulting and implementation of Data Quality frameworks, Data Catalog Metadata management, and Data Lineage.
- Excellent communication skills, self-motivated with a high degree of attention, can work independently.
- Experienced in building Automation Regressing Scripts for validation of ETL process between multiple databases like Oracle, SQL Server, Hive, and Mongo DB using Python.
- Good Experience in implementing and orchestrating data pipelines using Oozie and Airflow.
- Proficiency in SQL across several dialects (we commonly write MySQL, PostgreSQL, Redshift, Good knowledge of Data Marts, OLAP, Dimensional Data Modeling with Ralph Kimball Methodology (Star Schema Modeling, Snow-Flake Modeling for FACT and Dimensions Tables) using Analysis Services.
- Constructed data staging layers and fast real-time systems to feed BI applications and machine learning algorithms.

Professional Experience

Sr. Cloud Data Engineer, TIAA Financial Services

Oct 2024 – Present

Responsibilities: -

- Developed ETL pipelines using Azure Data Factory to extract, transform, and load data from various sources into Azure data stores.
- Implemented data security measures to ensure compliance with regulations such as GDPR and HIPAA.
- Collaborated with cross-functional teams to gather requirements and deliver scalable data solutions.
- Implemented data orchestration and scheduling using Azure Data Factory pipelines.
- Built and optimized data models in Azure SQL Database and Azure Synapse Analytics. Worked on AWS Data pipeline to configure data loads from S3 to Redshift.
- Extracted, transformed, and loaded data from various heterogeneous data sources and destinations using AWS Redshift.
- Developed a python script to hit REST API's and extract data to AWS S3
- Created Tables, Tableau/ Power BI Stored Procedures, and extracted data using T-SQL for business users whenever required.
- Selected and generated data into csv files and stored them into AWS S3 by using AWS EC2 and then structured and stored in AWS Redshift.
- Utilized Spark SQL API in PySpark to extract and load data and perform SQL queries.
- Worked on developing Pyspark script to encrypt the raw data by using hashing algorithms concepts on client specified columns.
- Used PySpark and Pandas to calculate the moving average and RSI score of the stocks and generated them into data warehouse.
- Exploring with Spark to improve the performance and optimization of the existing algorithms in Hadoop using Spark context, Spark-SQL, PostgreSQL, Data Frame, OpenShift, Talend, pair RDD's.
- Involved in integration of Hadoop cluster with spark engine to perform BATCH and GRAPHX operations.
- Performed data preprocessing and feature engineering for further predictive analytics using Python Pandas.
- Responsible for Design, Development, and testing of the database and Developed Stored Procedures, Views, and Triggers
- Compiling and validating data from all departments and Presenting to Director Operation.
- Created Tableau reports with complex calculations and worked on Ad-hoc reporting using Tableau.
- Developed data pipeline using Spark, Hive, Pig, python, Impala, and HBase to ingest customers.
- Involved in converting Hive/SQL queries into Spark transformations using Spark RDDs, Python and Scala.

Sr. Data Engineer, Envision Healthcare Remote, TX

May 2023 – Aug 2024

Responsibilities: -

- Involved in writing Spark applications using Python to perform various data cleansing, validation, transformation, and summarization activities according to the requirement.
- Creating pipelines in ADF employing linked services, datasets, and pipelines to extract, transform, and load data from many sources, such as Azure SQL, Blob storage, Azure SQL Data warehouse, FTP/SFTP, S3, and Netezza.
- Gathering requirements, designing, developing, and analyzing data models are all part of the processing.
- Conducted research on development and designing of sample methodologies like Waterfall.

- Migration of on-premises databases to Azure Cloud via Spark Streaming and batch processes, Azure Pipelines, Azure Databricks, Azure CICD, and Azure Data Lake.
- Creating complex queries in Data Bricks using Advanced SQL.
- Worked closely with Business Clients and Data Analytics to gather requirements and deliver accurate data.
- Worked with Azure Dev-ops to set up and implemented CICD pipelines.
- We deployed and tested pipelines between the United States and India using Azure templates.
- Involved from Start to End processes of Converting complex SAS reports to Hadoop scheduled jobs.
- Worked on writing APIs to load the processed data to HBase tables.
- Created Data Quality Scripts using SQL and Hive to validate successful data load and quality of the data.
- Prior to ingesting the data into HBase, we use Spark to process it and used it to create both batch and real-time spark jobs.
- Implemented Kafka producer and consumer applications on a Kafka cluster.
- Practical understanding of the Data modeling (Dimensional & Relational) concepts like Star-Schema Modeling, Snowflake Schema Modeling, Fact and Dimension tables.
- Handled Google Cloud Storage to use the storage service efficiently.
- Optimization of Hive queries using best practices and right parameters and using technologies like Hadoop, YARN, Python, PySpark.
- Participated in the migration of Teradata queries to Snowflake Data Warehouse queries.
- Work with Engineers, Scrum Masters, and Product Owners in a dynamic and exciting agile environment.

Data Engineer, Graks Info, Hyderabad, India

Apr 2019 – Dec 2022

Responsibilities: -

- Will be responsible for developing a highly scalable and flexible authority engine for all customer data.
- Developed a framework for converting existing PowerCenter mappings and to PySpark (Python and Spark) Jobs.
- Coordinating with development teams to determine application requirements and writing scalable code using Python programming language.
- Defined virtual warehouse sizing for Snowflake for a different type of workloads.
- Developed SQL queries using Snow SQL. Optimized and fine-tuned those queries. Work alongside AWS architecture and engineering teams to design and implement any scalable software services
- ETL pipelines in and out of data warehouse using a combination of Python and Snowflakes Snow SQL Writing SQL queries against Snowflake.
- Setup Alerting and monitoring using stack driver in GCP.
- Loading data from Linux/Unix file system to HDFS and working with PUTTY for better communication between Unix and Window system and for accessing the data files in the Hadoop environment.
- Developed and implemented HBase capabilities for Big de-normalized data set and then apply the transformation on the de-normalized data set using Spark/Scala.
- Created Spark Jobs to extract data from Hive tables and process the same using Data proc. Historical data load to Cloud Storage using Hadoop utilities and load to Big Query using BQ tools
- Worked on Google Cloud Platform Architecture for business needs by designing, building, and configuring applications to meet business process and application requirements.
- Developed and Implemented Spark ETL custom component to extract the data from upstream systems and push the data to HDFS and finally, store the data in HBase with wide row format.
- Creating and cloning the jobs and Job streams in the TWS tool and promoting them to higher environments.
- Design and develop data pipeline architectures using Hadoop, Spark, and related AWS Services.
- Conducted performance tuning and optimization of data processes to improve efficiency.

Hadoop Engineer, Wipro LTD, Hyderabad, India

Jun 2018 – Mar 2019

Responsibilities: -

- Developed Hive Scripts, Hive UDFs, and Python Scripting and used Spark (Spark-SQL, Spark-shell) to process data in Hortonworks.
- Performed advanced procedures like text analytics and processing using the in-memory computing capabilities of Spark.
- Usage of Sqoop to import data into HDFS from MySQL database and vice versa.
- Implemented optimized joins to perform analysis on different data sets using MapReduce programs.
- Created continuous integration and continuous delivery (CI/CD) pipeline on AWS that helps to automate steps in software delivery process.
- Experience in processing load and transforming the large data sets of structured, unstructured and semi structured data in Hortonworks.
- Implemented Partitioning, Dynamic Partitions and Buckets in HIVE & Impala for efficient data access.
- Configured Spark streaming (receivers) to receive Kafka input streams from the Kafka and Specified exact block interval for data Processing into HDFS using Scala.
- Collect the data using Spark streaming and dump into HBase and Cassandra. Used the Spark- Cassandra Connector to load data to and from Cassandra.
- Collecting and aggregating large amounts of log data using Kafka and staging data in HDFS Data Lake for further analysis.

Technical Skills

- **BigData/Hadoop Technologies:** MapReduce, Spark, Cloudera, Spark SQL, Azure HD Insights, Impala, AWS, Spark Streaming, Kafka, PySpark, Pig, Hive, HBase, Flume, Flink, Yarn
- **Languages:** HTML5, DHTML, WSDL, CSS3, C, C++, XML, R/R Studio, SAS Enterprise Guide, SAS, R (Caret, Weka, ggplot), Perl, MATLAB, Mathematica, FORTRAN, DTD, Schemas, Json, Ajax, Java, Scala, Python (NumPy)
- **NOSQL Databases:** Cassandra, HBase, MongoDB, MariaDB, Oracle, MySql, SQL Server
- **Web Design Tools:** HTML, CSS, JavaScript, JSP, jQuery, XML, JDBC, Struts
- **Development Tools:** Microsoft SQL Studio, IntelliJ, Azure Databricks, Eclipse, NetBeans
- **Public Cloud:** EC2, IAM, S3, Autoscaling, CloudWatch, Route53, EMR, RedShift, Glu
- **Orchestration tools:** Oozie, Airflow
- **Build Tools:** Jenkins, Toad, SQL Loader, PostgreSQL, Talend, Maven, ANT, RTC, RSA
- **Databases:** Microsoft SQL Server 2008,2010/2012, MySQL 4.x/5.x, Oracle 11g, 12c

Education

- **Master of Science in Computer Science**
Campbellsville University, USA
- **Bachelor of Commerce in Computer**
AISECT University, India